

Midterm Exam 1 Solutions

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

True/False

1. In BPSK over AWGN with equal priors, the ML and MAP decision rules yield the same decision threshold. **True** - equal priors make ML = MAP, threshold at 0 (after whitening).
2. The thermal noise power at room temperature scales linearly with both absolute bandwidth and temperature. **True** - $P_n = kTB$.
3. For M -QAM, SER is always strictly lower than BER at the same E_b/N_0 . **False** - typically $\text{BER} \leq \text{SER}$ (with Gray coding, $\text{BER} \approx \text{SER} / \log_2 M$ at high SNR).
4. Ideal bandlimited sampling at $f_s > 2W$ permits perfect reconstruction using an ideal low-pass filter. **True** - Shannon-Nyquist criterion.
5. In a narrowband (flat) fading channel, the multipath channel can be well-approximated by a single complex gain. **True** - frequency response is flat over the signal band.
6. Inter-symbol interference (ISI) becomes more severe as the symbol time approaches the delay spread of the channel. **True** - less guard relative to multipath spread increases overlap.
7. For a real passband signal, the spectrum is conjugate-symmetric; for complex baseband, it need not be. **True** - Hermitian symmetry only for real signals.
8. Frequency-selective fading can occur even when the overall signal bandwidth is very small compared to the channel's coherence bandwidth. **True/False, both correct** - if $B \ll B_c$, fading is (approximately) flat, not frequency-selective.
9. In OFDM, orthogonality allows tightly packed subcarriers without mutual interference at the sampling instants. **True** - sinc zeros at other subcarriers (with perfect sync/CP).
10. Carrier frequency offset (CFO) causes a common phase rotation across subcarriers that accumulates over OFDM symbols. **True** - induces symbol-to-symbol phase drift (and ICI if nonzero).

Multiple Choice Questions

1. (b)/(d) - Equal priors $\Rightarrow$ ML = MAP; threshold at 0.
2. (d) - MAP boundary shifts toward the *less likely* symbol (to favor the more likely one).

3. (c) - For fixed E_b/N_0 , lower-order constellations (QPSK) have lower BER.
4. (c) - Nyquist-Shannon: perfect reconstruction with $f_s \geq 2W$ and ideal LPF.
5. (a) - Flat fading $\Rightarrow H(f)$ approximately constant over the signal band.
6. (b) - No ISI if CP exceeds the channel's maximum excess delay: $T_{\text{CP}} > \max_k \tau_k$.
7. (a) - CP maintains subcarrier orthogonality under multipath; not for efficiency/PAPR/noise.
8. (c) - Severe frequency selectivity when signal bandwidth $\gg$ coherence bandwidth.
9. (c) - Residual CFO causes a common phase that drifts over OFDM symbols.
10. (d) - OFDM partitions a frequency-selective channel into many flat subchannels.

Question 1: Modulation, Sampling and Channel Effects

(a) Minimum Euclidean distance for 4-QAM (QPSK).

For a standard square 4-QAM with points at $(\pm a, \pm a)$, the average symbol energy is

$$E_s = \mathbb{E}\{|s|^2\} = 2a^2 \Rightarrow a = \sqrt{\frac{E_s}{2}}.$$

Adjacent symbols differ by $2a$ along one axis, so

$$d_{\min} = 2a = 2\sqrt{\frac{E_s}{2}} = \boxed{\sqrt{2E_s}}.$$

(b) Aliasing when $B = 3$ MHz and $f_s = 4$ MHz.

The continuous-time spectrum $X(f)$ is nonzero on $f \in [-3, 3]$ MHz. After ideal sampling at $f_s = 4$ MHz, the discrete-time spectrum consists of replicas $X(f - kf_s)$ for all integers k :

Replicas centered at $f = kf_s$, each spanning $[kf_s - 3, kf_s + 3]$ MHz.

The first replicas ($k = \pm 1$) occupy:

$$[1, 7] \text{ MHz and } [-7, -1] \text{ MHz.}$$

Since the baseband copy is on $[-3, 3]$ MHz, overlaps occur in

$$[1, 3] \text{ MHz and } [-3, -1] \text{ MHz,}$$

i.e., *aliasing* in the outer portions of the baseband. The only alias-free baseband region is

$$|f| < 1 \text{ MHz.}$$

Qualitative sketch: draw the original lowpass lobe on $[-3, 3]$ MHz; then plot shifted copies centered at ± 4 MHz; highlight the overlap (folding) in $[1, 3]$ and $[-3, -1]$ MHz.

(c) Two-path phase difference at $f_c = 5$ GHz.

Wavelength: $\lambda = \frac{c}{f_c} = \frac{3 \times 10^8}{5 \times 10^9} = 0.06$ m. Path-length difference: $\Delta d = d_2 - d_1 = 10.06 - 10 = 0.06$ m $= \lambda$. Phase difference:

$$\Delta\phi = 2\pi \frac{\Delta d}{\lambda} = 2\pi \frac{0.06}{0.06} = \boxed{2\pi \text{ rad } (= 360^\circ)}.$$

Thus the paths are **in phase** and the interference is **constructive**.

Question 2: BER of OOK

Given: $y = s + v$, $s \in \{0, A\}$, $v \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = N_0/2$. Let $Q(x) \triangleq \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$.

- (a) **Decision boundary (intuition, equal priors).**

Place the threshold halfway between the two symbols:

$$\boxed{y \underset{\hat{s}=0}{\overset{\hat{s}=A}{\gtrless}} \gamma^*, \quad \gamma^* = \frac{A}{2}}.$$

- (b) **Average BER with equal priors.**

$$P_b = \frac{1}{2} \Pr(y > \gamma^* | s = 0) + \frac{1}{2} \Pr(y < \gamma^* | s = A) = \frac{1}{2} Q\left(\frac{\gamma^*}{\sigma}\right) + \frac{1}{2} Q\left(\frac{A-\gamma^*}{\sigma}\right).$$

With $\gamma^* = A/2$ the two terms are equal, hence

$$\boxed{P_b = Q\left(\frac{A}{2\sigma}\right)}.$$

- (c) **Relate A and E_b (unit-energy pulse, 1 bit/symbol).**

Only the 'on' symbol carries energy; with equiprobable bits

$$E_b = \frac{1}{2} A^2 \implies \boxed{A = \sqrt{2E_b}}.$$

- (d) **Express P_b in E_b/N_0 .**

Using $\sigma^2 = N_0/2$,

$$\frac{A}{2\sigma} = \frac{\sqrt{2E_b}}{2\sqrt{N_0/2}} = \sqrt{\frac{E_b}{N_0}},$$

so

$$\boxed{P_b = Q\left(\sqrt{E_b/N_0}\right)}$$

- (e) **Comparison to coherent BPSK.**

BPSK has $P_b^{\text{BPSK}} = Q\left(\sqrt{2E_b/N_0}\right)$, so OOK's Q -argument is smaller by a factor $\sqrt{2}$, i.e.,

$$\boxed{\text{OOK requires } \approx 3 \text{ dB more } E_b/N_0 \text{ than BPSK for the same BER.}}$$

Reason: OOK transmits no energy for the "0" symbol, reducing the minimum distance between hypotheses (and the effective detection margin) compared to BPSK.

Question 3: Multipath Channel

Given: Complex baseband $x_b(t)$, analytic/complex passband $x_p(t) = x_b(t) e^{j2\pi f_c t}$. Multipath passband I/O:

$$y_p(t) = \sum_{k=1}^K h_k x_p(t - \tau_k) + v_p(t),$$

with complex path gains h_k and delays τ_k . Define the IQ-mixer output (ideal LO, no filtering drawn) as

$$y_b(t) \triangleq y_p(t) e^{-j2\pi f_c t}.$$

Derivation:

$$\begin{aligned}
y_b(t) &= \left[\sum_{k=1}^K h_k x_p(t - \tau_k) + v_p(t) \right] e^{-j2\pi f_c t} \\
&= \sum_{k=1}^K h_k \underbrace{x_p(t - \tau_k)}_{x_b(t - \tau_k) e^{j2\pi f_c (t - \tau_k)}} e^{-j2\pi f_c t} + v_p(t) e^{-j2\pi f_c t} \\
&= \sum_{k=1}^K h_k x_b(t - \tau_k) e^{j2\pi f_c (t - \tau_k)} e^{-j2\pi f_c t} + w(t) \\
&= \sum_{k=1}^K h_k x_b(t - \tau_k) e^{-j2\pi f_c \tau_k} + w(t),
\end{aligned}$$

where we defined the complex baseband noise $w(t) \triangleq v_p(t) e^{-j2\pi f_c t}$.

Result:

$$y_b(t) = \sum_{k=1}^K \underbrace{h_k e^{-j2\pi f_c \tau_k}}_{\text{baseband tap}} x_b(t - \tau_k) + w(t)$$

Remark: Using an analytic/complex passband $x_p(t)$ eliminates the $\pm 2f_c$ image terms. If a *real* passband $s_p(t) = \Re\{x_b(t) e^{j2\pi f_c t}\}$ were used, multiplying by $e^{-j2\pi f_c t}$ creates a desired term as above *plus* an image centered at $\pm 2f_c$, which is then conceptually discarded (ideal low-pass).

Question 4: OFDM System Design

Given: Occupied bandwidth $B = 20$ MHz, subcarrier spacing $\Delta f = 15$ kHz (later 30 kHz), QPSK (2 bits/subcarrier), no pilots/guards/coding. Use a power-of-two N_{FFT} , but only the subcarriers within B are populated. Start with no CP, then add CP to cover $T_d = 4 \mu\text{s}$. Ignore other overheads.

(a) **Needed subcarriers and N_{FFT} .**

$$N_{\text{req}} = \frac{B}{\Delta f} = \frac{20 \cdot 10^6}{15 \cdot 10^3} = 1333.\bar{3}.$$

Number of *occupied/data* subcarriers (not exceeding B):

$$N_{\text{occ}} = \lfloor 1333.\bar{3} \rfloor = 1333.$$

Choose the next power of two for efficient FFT:

$$N_{\text{FFT}} = 2048.$$

Sampling rate (used later for CP samples):

$$F_s = N_{\text{FFT}} \Delta f = 2048 \cdot 15 \text{ kHz} = 30.72 \text{ MHz}.$$

(b) **Useful OFDM symbol duration.**

$$T_u = \frac{1}{\Delta f} = \frac{1}{15 \text{ kHz}} = 66.6667 \text{ } \mu\text{s}.$$

(c) **Throughput with no CP (QPSK on N_{occ}).**

$$\text{bits/symbol} = N_{\text{occ}} \cdot 2 = 1333 \cdot 2 = \boxed{2666 \text{ bits}}, \quad \text{symbols/s} = \frac{1}{T_u} = 15,000 \text{ s}^{-1}.$$

$$R_{\text{no-CP}} = 2666 \times 15,000 = 39.99 \text{ Mbit/s}.$$

Question 5 (ECSE 6560 Only): Residual CFO and SFO

Given:

$$B = 20 \text{ MHz}, F_s = 20 \text{ MHz}, T_s = \frac{1}{F_s} = 50 \text{ ns}, N = 64, N_{\text{CP}} = 16, N + N_{\text{CP}} = 80.$$

Per-symbol duration (incl. CP): $T_{\text{sym}} = (N + N_{\text{CP}}) T_{\text{sample}} = 80 \cdot 50 \text{ ns} = 4 \text{ } \mu\text{s}$. Pilot regression: $m = 0.05 \text{ rad/index}$, $b = -0.07 \text{ rad}$.

Phase-tracking relation (index domain).

$$\Delta\phi(f) = \underbrace{2\pi f \frac{(N + N_{\text{CP}})}{N} \delta T_s}_{\text{SFO (slope term)}} - \underbrace{2\pi \delta f_c (N + N_{\text{CP}}) T_s}_{\text{CFO (intercept term)}} = mf + b,$$

so

$$m = 2\pi \frac{(N + N_{\text{CP}})}{N} \delta T_s, \quad b = -2\pi \delta f_c (N + N_{\text{CP}}) T_{\text{sample}}.$$

They derive residual CFO $\delta f_c = 2.8 \text{ kHz}$ and SFO time offset $\delta T_s = 0.0064 \text{ } \mu\text{s}$.

(a) **Cumulative constellation rotation from CFO over 100 symbols.**

Per-symbol rotation (CFO-only):

$$\Delta\phi_{\text{sym}} = -2\pi \delta f_c T_{\text{sym}} = -2\pi \cdot (2.8 \text{ kHz}) \cdot (4 \text{ } \mu\text{s}) = -0.0224\pi \approx -0.07 \text{ rad} \approx -4.01^\circ.$$

Over 100 symbols:

$$\theta_{100} = -100 \cdot \Delta\phi_{\text{sym}} = -2.24\pi \approx -7 \text{ rad} \approx -401^\circ.$$

This is more than one full revolution, so the rotation would be clearly visible in a constellation plot.

(b) **Doubling to $N = 128$ at the same sampling rate.**

Assume $F_s = 20 \text{ MHz}$ unchanged and *keep the same CP length in samples* $N_{\text{CP}} = 16$ (so $T_{\text{CP}} = 0.8 \text{ } \mu\text{s}$ unchanged). Then:

$$N' = 128, \quad N'_{\text{CP}} = 16, \quad (N' + N'_{\text{CP}}) = 144, \quad T'_{\text{sym}} = 144 \cdot 50 \text{ ns} = 7.2 \text{ } \mu\text{s}.$$

CFO impact per symbol.

$$\Delta\phi'_{\text{sym}} = 2\pi \delta f_c T'_{\text{sym}} = \Delta\phi_{\text{sym}} \cdot \frac{T'_{\text{sym}}}{T_{\text{sym}}} = \Delta\phi_{\text{sym}} \cdot \frac{7.2}{4.0}, \boxed{\text{(about } 1.8 \times \text{ larger per symbol)}}.$$

SFO impact per subcarrier index. Slope factor changes from $\frac{N+N_{\text{CP}}}{N} = \frac{80}{64} = 1.25$ to $\frac{N'+N'_{\text{CP}}}{N'} = \frac{144}{128} = 1.125$. For the *same* δT_s (fraction of T_s), the per-index SFO-induced phase slope becomes

$$m' = 2\pi \frac{(N' + N'_{\text{CP}})}{N'} \delta T_s = m \cdot \frac{1.125}{1.25} = \boxed{0.9 m \text{ (10\% smaller per index)}}.$$

Intuitively: the longer symbol increases CFO accumulation per symbol (linear in T_{sym}), while the index-domain SFO slope shrinks slightly because $(N + N_{\text{CP}})/N$ decreases when N increases with CP held fixed in samples.